

T-Distribution Explained

MDragon Data Tools - Study Guide

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1. What is the T-Distribution?

The **t-distribution** (also called Student's t-distribution) is a probability distribution used when:

- Working with small sample sizes ($n < 30$)
- Population standard deviation (σ) is unknown
- Estimating population parameters from sample data

Who Uses It?

- Researchers with limited sample sizes
- Anyone creating confidence intervals
- Hypothesis testing with unknown σ
- Quality control with small batches

2. T-Distribution vs Normal Distribution

Similarities

- Both are symmetric and bell-shaped
- Both centered at 0
- Both used for inference about means

Key Differences

Feature	Normal Distribution	T-Distribution
Shape	Fixed shape	Varies with df
Tails	Thinner tails	Heavier/fatter tails
When Used	σ known, large n	σ unknown, small n
Parameters	μ and σ	Degrees of freedom (df)
As n grows	Stays same	Approaches normal

Visual Comparison

Normal (thin tails)



T-distribution (fat tails)



← More probability in tails

3. Degrees of Freedom (df)

What is Degrees of Freedom?

Degrees of freedom represents the number of independent pieces of information available to estimate a parameter.

Formula

For a single sample:

$$df = n - 1$$

where n = sample size

For two samples (pooled):

$$df = n_1 + n_2 - 2$$

Why df Matters

- **Smaller df** → Heavier tails → More uncertainty
- **Larger df** → Approaches normal distribution
- **df = 30+** → Nearly identical to normal

Example df Values

Sample Size (n)	df	Shape
5	4	Very heavy tails
10	9	Heavy tails
20	19	Moderate tails
30	29	Nearly normal
100	99	Essentially normal

4. When to Use T-Distribution

Use T-Distribution When:

1. **Small sample size** ($n < 30$)
2. **Population σ is unknown** (use sample s instead)
3. **Assuming normality** (or approximately normal data)

Use Normal Distribution When:

1. Large sample size ($n \geq 30$) - Central Limit Theorem applies
2. Population σ is known
3. Working with proportions

Decision Flowchart

Is σ known?

|

├ Yes → Use Normal (Z)

|

└ No → Is $n \geq 30$?

|

├ Yes → Can use Normal (Z) or T (similar results)

|

└ No → Use T-distribution

5. Step-by-Step Examples

Example 1: Finding a T-Score

 **Problem:** A sample of $n = 15$ students has a mean test score of $\bar{x} = 82$ with $s = 8$. The population mean is hypothesized to be $\mu = 75$. Calculate the t-score.

 **Solution:**

1. Identify the values:

- $\bar{x} = 82$ (sample mean)
- $\mu = 75$ (hypothesized population mean)
- $s = 8$ (sample standard deviation)
- $n = 15$ (sample size)

2. Calculate degrees of freedom:

$$df = n - 1 = 15 - 1 = 14$$

3. Calculate t-score:

$$\begin{aligned} t &= (\bar{x} - \mu) / (s / \sqrt{n}) \\ &= (82 - 75) / (8 / \sqrt{15}) \\ &= 7 / (8 / 3.873) \\ &= 7 / 2.066 \\ &= 3.39 \end{aligned}$$

✓ Answer: $t = 3.39$ with $df = 14$

Interpretation: The sample mean is 3.39 standard errors above the hypothesized population mean.

Example 2: Critical T-Value for Confidence Interval

 **Problem:** Find the critical t-value for a 95% confidence interval with $n = 20$.

 **Solution:**

1. Calculate degrees of freedom:

$$df = n - 1 = 20 - 1 = 19$$

2. Determine significance level:

$$\begin{aligned} \text{Confidence level} &= 95\% = 0.95 \\ \alpha &= 1 - 0.95 = 0.05 \\ \alpha/2 &= 0.025 \text{ (two-tailed)} \end{aligned}$$

3. Look up t-table:

- $df = 19$
- $\alpha/2 = 0.025$
- $t\text{-critical} = 2.093$

✓ Answer: $t^* = \pm 2.093$

Interpretation: For a 95% CI with 19 df, use t-values of ± 2.093 .

Example 3: Constructing a Confidence Interval

 **Problem:** A sample of 12 measurements has $\bar{x} = 50$ and $s = 6$. Construct a 90% confidence interval for the population mean.

 **Solution:**

1. Identify values:

- $\bar{x} = 50$
- $s = 6$
- $n = 12$
- Confidence level = 90%

2. Calculate df:

$$df = 12 - 1 = 11$$

3. Find t-critical (90% CI, df = 11):

$$\begin{aligned}\alpha &= 0.10, \alpha/2 = 0.05 \\ t^* &= 1.796 \text{ (from t-table)}\end{aligned}$$

4. Calculate margin of error:

$$\begin{aligned}ME &= t^* \times (s / \sqrt{n}) \\ &= 1.796 \times (6 / \sqrt{12}) \\ &= 1.796 \times (6 / 3.464) \\ &= 1.796 \times 1.732 \\ &= 3.11\end{aligned}$$

5. Construct interval:

$$\begin{aligned} \text{CI} &= \bar{x} \pm \text{ME} \\ &= 50 \pm 3.11 \\ &= (46.89, 53.11) \end{aligned}$$

✓ **Answer:** 90% CI: (46.89, 53.11)

Interpretation: We are 90% confident the true population mean is between 46.89 and 53.11.

Example 4: Comparing T vs Z

 **Problem:** For a 95% confidence interval, compare the critical values for:

- a) $df = 5$
- b) $df = 30$
- c) Z-distribution (infinite df)

 **Solution:**

1. For 95% CI, $\alpha/2 = 0.025$
2. Look up values:

$$\begin{aligned} df = 5: & \quad t^* = 2.571 \\ df = 30: & \quad t^* = 2.042 \\ Z: & \quad z^* = 1.96 \end{aligned}$$

✓ **Answer:**

- Small sample ($df=5$): $t^* = 2.571$ (much larger)
- Moderate sample ($df=30$): $t^* = 2.042$ (close to Z)
- Large sample (Z): $z^* = 1.96$

Interpretation: As sample size increases, t-values approach z-values. Smaller samples require larger multipliers for the same confidence level.

Example 5: One-Sample T-Test

 **Problem:** A company claims their batteries last 100 hours on average. A sample of 8 batteries has $\bar{x} = 95$ hours and $s = 10$ hours. Test at $\alpha = 0.05$ if the true mean is less than 100.

 **Solution:**

1. Set up hypotheses:

$$\begin{aligned}H_0: \mu &= 100 \text{ (claim)} \\H_a: \mu &< 100 \text{ (left-tailed test)}\end{aligned}$$

2. Calculate test statistic:

$$\begin{aligned}df &= 8 - 1 = 7 \\t &= (95 - 100) / (10 / \sqrt{8}) \\&= -5 / (10 / 2.828) \\&= -5 / 3.536 \\&= -1.414\end{aligned}$$

3. Find critical value:

$$\begin{aligned}\alpha &= 0.05 \text{ (left-tailed), } df = 7 \\t\text{-critical} &= -1.895\end{aligned}$$

4. Make decision:

$$\begin{aligned}t &= -1.414 > t\text{-critical} = -1.895 \\&\text{Fail to reject } H_0\end{aligned}$$

 **Answer:** Do not reject the null hypothesis

Interpretation: There is insufficient evidence at the 0.05 level to conclude the batteries last less than 100 hours.

Example 6: Determining Sample Size Effect

 **Problem:** Compare the margin of error for a 95% CI with $s = 10$ for:

- a) $n = 10$
- b) $n = 25$

 **Solution:**

a) $n = 10$:

$$\begin{aligned}df &= 9, t^* = 2.262 \\ME &= 2.262 \times (10 / \sqrt{10}) \\&= 2.262 \times 3.162 \\&= 7.15\end{aligned}$$

b) $n = 25$:

$$\begin{aligned}df &= 24, t^* = 2.064 \\ME &= 2.064 \times (10 / \sqrt{25}) \\&= 2.064 \times 2 \\&= 4.13\end{aligned}$$

 **Answer:**

- $n = 10$: $ME = \pm 7.15$
- $n = 25$: $ME = \pm 4.13$

Interpretation: Increasing sample size from 10 to 25 reduces margin of error by about 42%, giving a more precise estimate.

6. Using T-Tables

How to Read a T-Table

1. Find your df (rows)
2. Find your α or confidence level (columns)

3. Locate the intersection - that's your t-critical value

Common Confidence Levels and α Values

Confidence Level	α	$\alpha/2$ (two-tailed)
90%	0.10	0.05
95%	0.05	0.025
98%	0.02	0.01
99%	0.01	0.005

Example T-Table Excerpt

df	$\alpha=0.10$	$\alpha=0.05$	$\alpha=0.025$	$\alpha=0.01$
5	1.476	2.015	2.571	3.365
10	1.372	1.812	2.228	2.764
15	1.341	1.753	2.131	2.602
20	1.325	1.725	2.086	2.528
30	1.310	1.697	2.042	2.457
∞	1.282	1.645	1.960	2.326 (Z-values)

7. Practice Problems

Problem 1

Calculate the t-score for a sample where:

- $\bar{x} = 65, \mu = 60, s = 12, n = 18$

Problem 2

Find the critical t-value for:

- a) 95% CI, $n = 25$
- b) 99% CI, $n = 10$

Problem 3

Construct a 95% confidence interval for μ given:

- $\bar{x} = 120$, $s = 15$, $n = 16$

Problem 4

A sample of 9 measurements has $\bar{x} = 48$ and $s = 8$. Test $H_0: \mu = 50$ vs $H_a: \mu \neq 50$ at $\alpha = 0.05$.

Problem 5

How does the margin of error change when going from:

- a) $n = 15$ to $n = 30$ (with $s = 5$, 95% CI)?

Answers to Practice Problems

Problem 1

$$\begin{aligned} df &= 18 - 1 = 17 \\ t &= (65 - 60) / (12 / \sqrt{18}) \\ &= 5 / 2.828 \\ &= 1.77 \end{aligned}$$

Problem 2

- a) 95% CI, $n = 25$:

$$df = 24, \alpha/2 = 0.025$$

$$t^* = 2.064$$

b) 99% CI, $n = 10$:

$$df = 9, \alpha/2 = 0.005$$

$$t^* = 3.250$$

Problem 3

$$df = 15, t^* = 2.131$$

$$ME = 2.131 \times (15 / \sqrt{16}) = 2.131 \times 3.75 = 7.99$$

$$CI = 120 \pm 7.99 = (112.01, 127.99)$$

Problem 4

$$df = 8, \alpha = 0.05 \text{ (two-tailed)}$$

$$t = (48 - 50) / (8 / \sqrt{9}) = -2 / 2.667 = -0.75$$

$$t\text{-critical} = \pm 2.306$$

$$|t| < t\text{-critical}, \text{ so fail to reject } H_0$$

Problem 5

$n = 15$:

$$df = 14, t^* = 2.145$$

$$ME = 2.145 \times (5 / \sqrt{15}) = 2.77$$

$n = 30$:

$$\begin{aligned}df &= 29, \quad t^* = 2.045 \\ME &= 2.045 \times (5 / \sqrt{30}) = 1.87\end{aligned}$$

Reduction: $(2.77 - 1.87) / 2.77 = 32.5\%$ smaller

Quick Reference

T-Score Formula

$$t = (\bar{x} - \mu) / (s / \sqrt{n})$$

Degrees of Freedom

$$df = n - 1 \text{ (single sample)}$$

Confidence Interval

$$CI = \bar{x} \pm t^* \times (s / \sqrt{n})$$

When to Use

- $n < 30$ AND σ unknown $\rightarrow$ Use T
- $n \geq 30$ OR σ known $\rightarrow$ Can use Z

Use our Normal Distribution Calculator at MDragon Data Tools - it handles both Z and T distributions!

MDragon Data Tools | Statistics Education Platform

mondragon-developer.github.io/statools